

Section 3: Organic chemistry

- a) Introduction
- b) Alkanes
- c) Alkenes
- d) Ethanol

Introduction

Students will be assessed on their ability to:

- 3.1 explain the terms homologous series, hydrocarbon, saturated, unsaturated, general formula and isomerism

b) Alkanes

Students will be assessed on their ability to:

- 3.2 recall that alkanes have the general formula C_nH_{2n+2}
- 3.3 draw displayed formulae for alkanes with up to five carbon atoms in a molecule, and name the straight-chain isomers
- 3.4 recall the products of the complete and incomplete combustion of alkanes
- 3.5 recall the reaction of methane with bromine to form bromomethane in the presence of UV light

c) Alkenes

Students will be assessed on their ability to:

- 3.6 recall that alkenes have the general formula C_nH_{2n}
- 3.7 draw displayed formulae for alkenes with up to four carbon atoms in a molecule, and name the straight-chain isomers
- 3.8 describe the addition reaction of alkenes with bromine, including the decolourising of bromine water as a test for alkenes

d) Ethanol

Students will be assessed on their ability to:

- 3.9 describe the **manufacture** of ethanol by passing ethene and steam over a phosphoric acid catalyst at a temperature of about 300°C and a pressure of about 60–70 atm
- 3.10 describe the **manufacture** of ethanol by the fermentation of sugars, for example glucose, at a temperature of about 30°C
- 3.11 evaluate the factors relevant to the choice of method used in the manufacture of ethanol, for example the relative availability of sugar cane and crude oil
- 3.12 describe the dehydration of ethanol to ethene, using aluminium oxide